

Instructions



How is it going ? Do you have problems with practical tasks ? Any concerns with learning materials ?

should we create erd

everything is good

scared all good gonna sleep

everything good

jpa task can't got it

Today's agenda

- Total Recall
- Database normalization
- Isolation
- Transaction
- Query
- Quiz
- Questions

Total Recall



What do you remember from the last session?

some annotations
entity jpa service
hibernate

What is *detached* state? vs *transient* ? vs *remove*?

transient - new object

removed pending deletion removed - delete

remove - remove detached - not observed

detached has database id

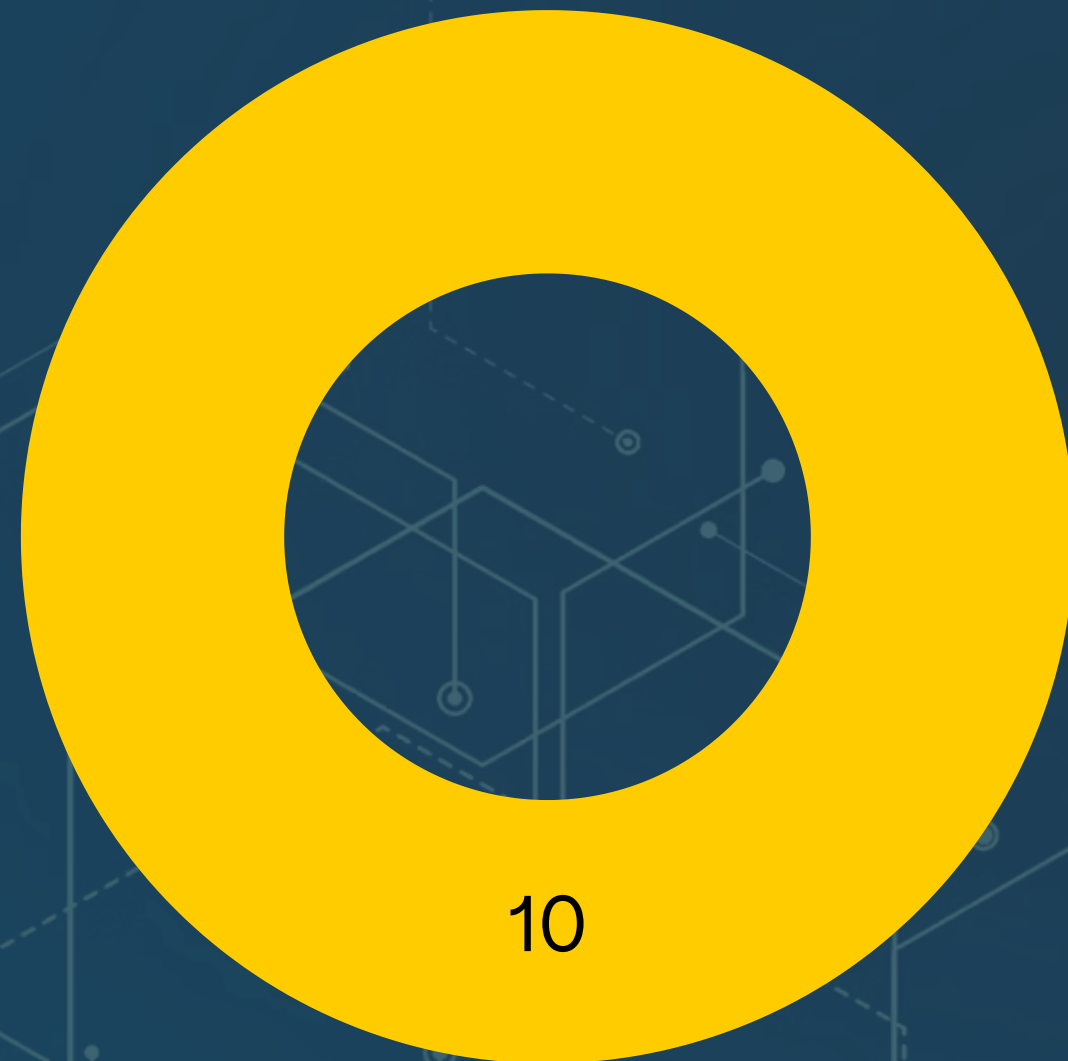
detached managed before transient - observed

detached - saved before

states of an entity idk

transient new untracked

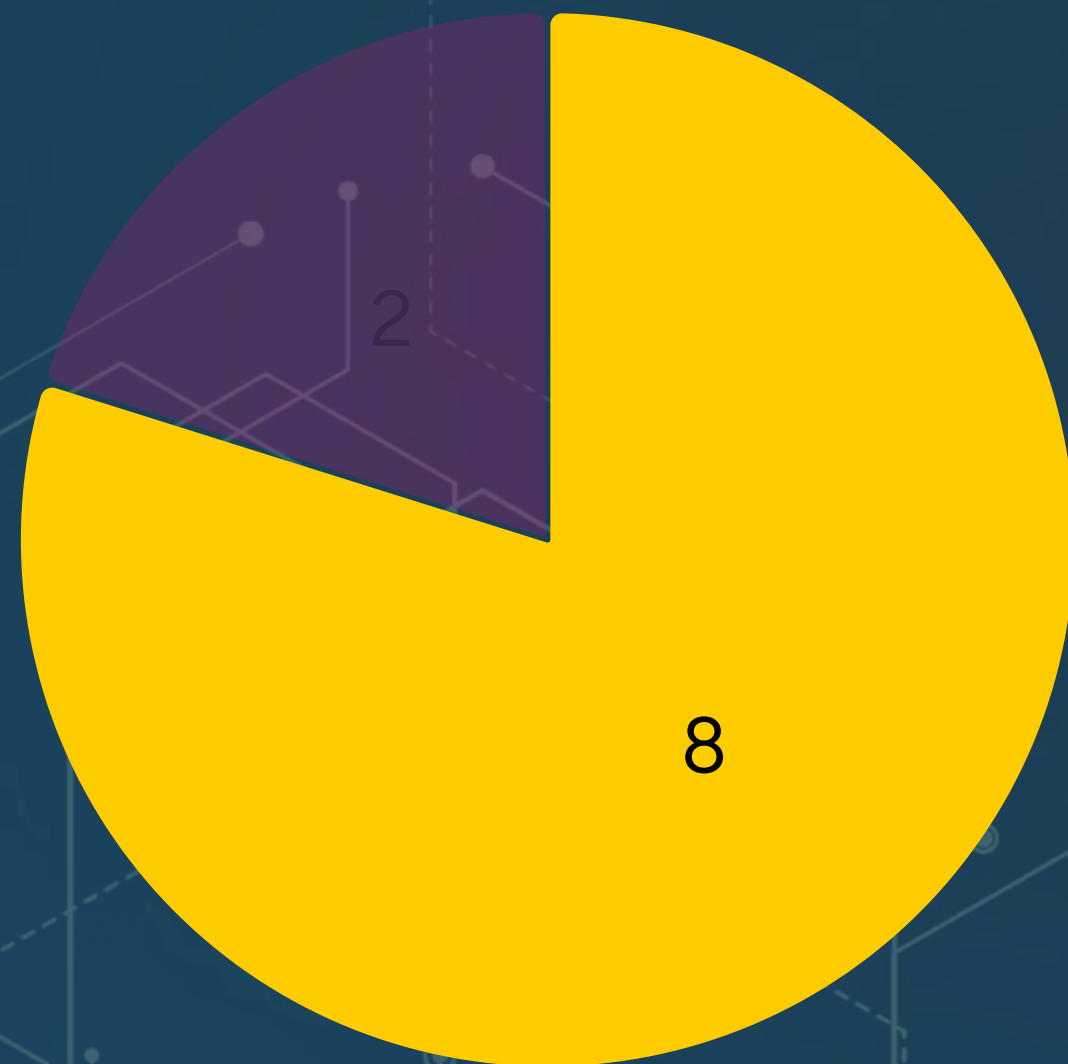
What does FetchType.LAZY do in JPA?



- 0 Forces to eagerly load associated entities
- 10 Allows to delay loading of associated entities until they are accessed
- 0 Automatically saves associated entities when parent entity is saved
- 0 Deletes associated entities when parent entity is deleted

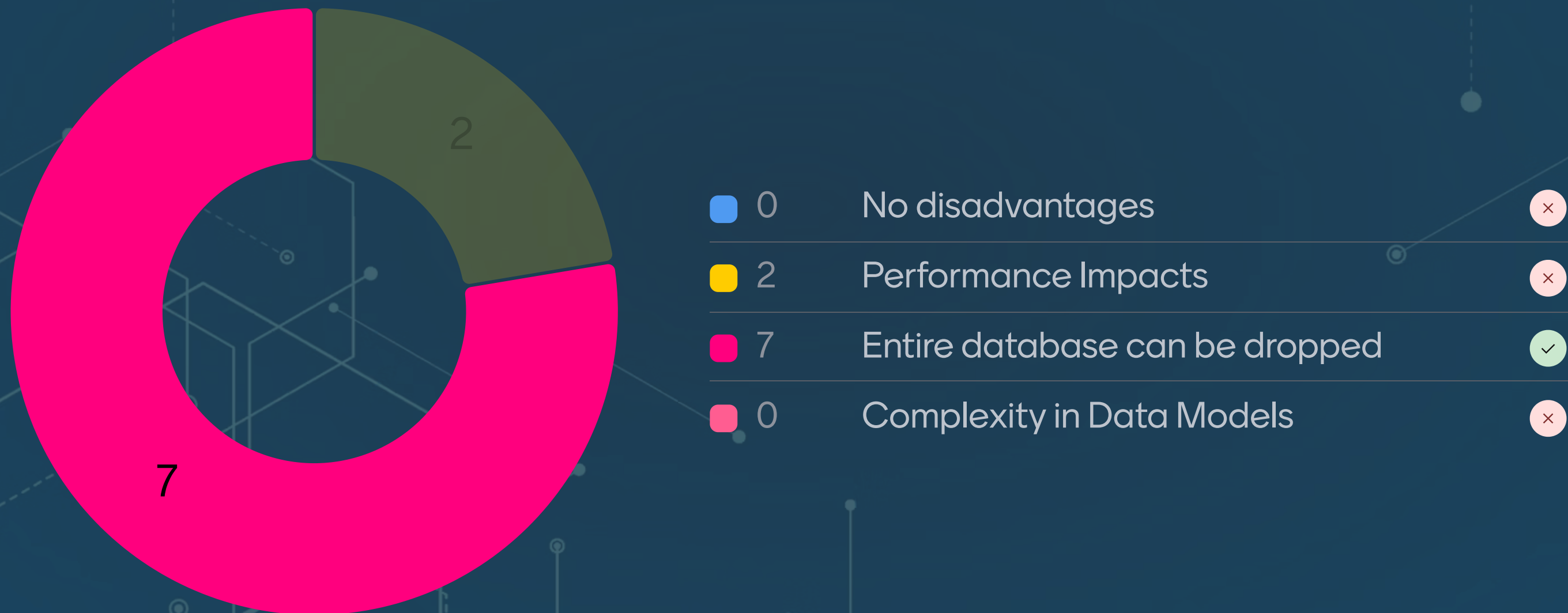


What's the purpose of *orphanRemoval* flag?

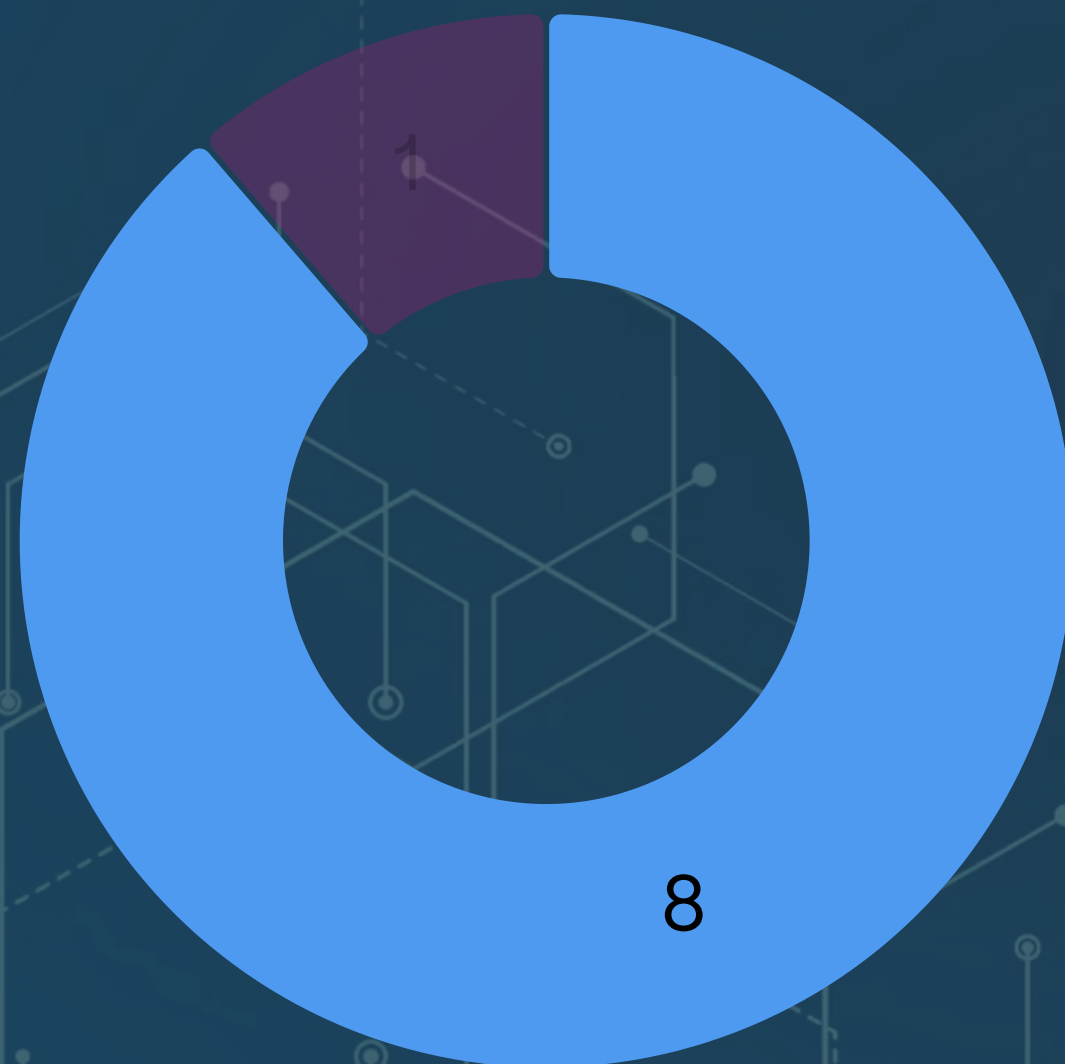


- ☐ 0 Same as Cascade Delete ☐
- ☒ 8 Child entity can not exist without Parent, and should be deleted ☒
- ☐ 2 When Child entity is deleted, then Parent Entity would be deleted ☐

What's the main disadvantage of using *CascadeType.REMOVE*?



What annotation(s) are used to setup a *compound* key?



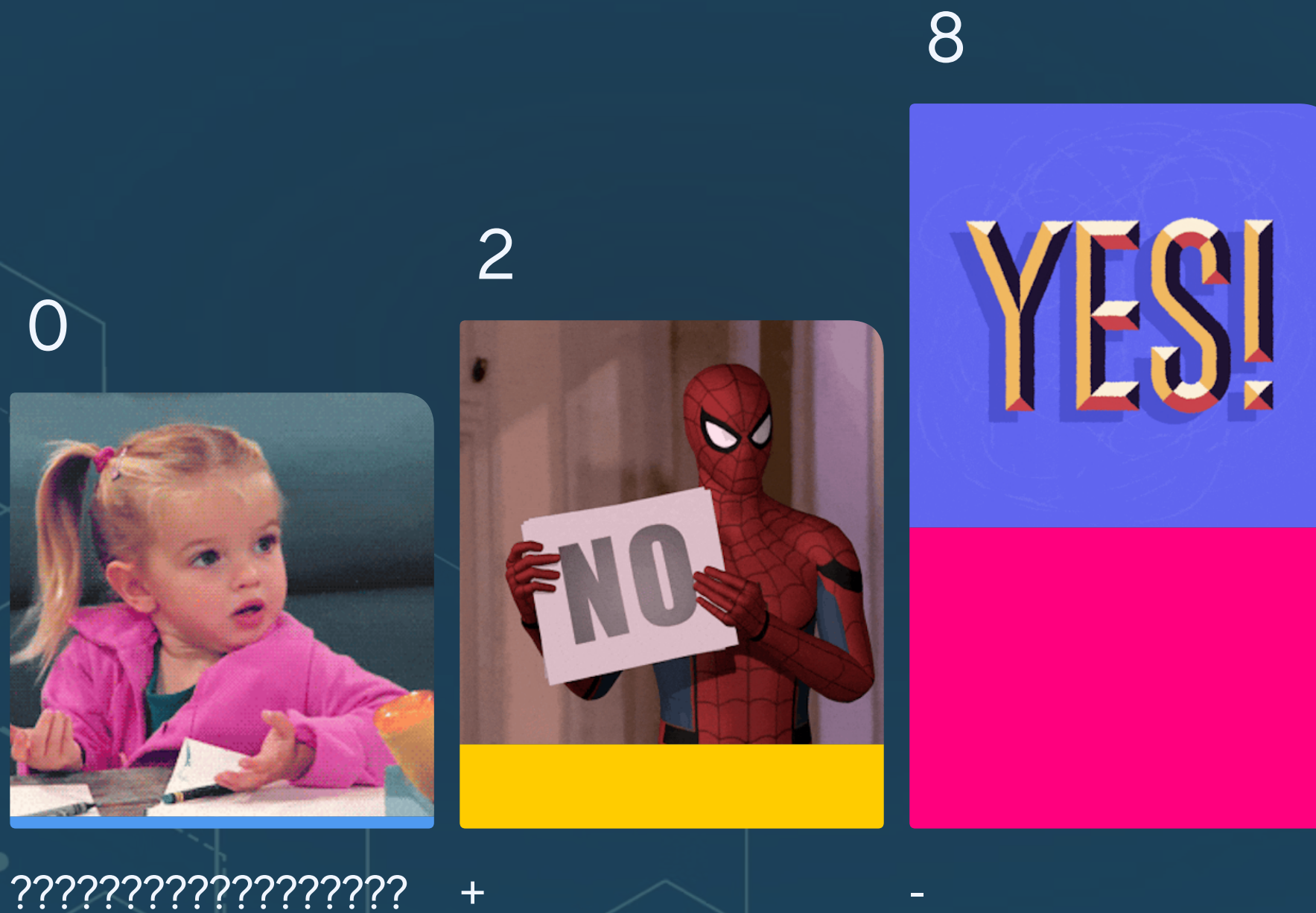
8	@EmbeddedId and @Embeddable	✓
0	It is yet impossible in JPA	✗
1	@CompoundId and @CompoundKey	✗

JPA (ws-06)

Normalization | Isolation | Transaction | Query



Have you started the theory of 'JPA'?



Database normalization

https://en.wikipedia.org/wiki/Database_normalization

https://en.wikipedia.org/wiki/Third_normal_form

ACID

<https://en.wikipedia.org/wiki/ACID>

Isolation

[https://en.wikipedia.org/wiki/Isolation_\(data
base_systems\)](https://en.wikipedia.org/wiki/Isolation_(database_systems)).

Explore module ws-06

<https://github.com/AndreiRohau/itpu-web-development-with-java-2026>

10 mins beak

Quiz

How do you begin a transaction in JPA?

19 ✓



0 ✗

`entityManager.begin();`

2 ✗

`entityTransaction.start();`

1 ✗

`entityManager.start();`

`entityTransaction.begin();`

In JPA, what annotation is used to rollback a transaction when a specific exception occurs?

20 ✓



4 ✗



@Rollback(Exception.class)

@Transactional(rollbackFor =
Exception.class)

0 ✗

@TransactionType(ROLLBACK)

0 ✗

@ManageTransaction(Exception
= "rollback")

What is TRUE about @Transactional annotation in JPA?

21 ✓

2 ✗

It ensures that all operations within method are committed to database

1 ✗

It auto creates new instances of entities in persistence context

0 ✗

Enforces to flush changes to database every time a query is executed

What does JPQL stand for?

23 ✓



0 ✗

Java Persistence Query Language

Java Predefined Query Language

1 ✗

Java Primary Query Language

0 ✗

Java Portable Query Language

What approach allows you to use database-specific features in JPA?

22 ✓



2 ✗

JPQL

0 ✗

Criteria API

Native Query

Which JPA API is used for type-safe queries?

23 ✓



Criteria API

1 ✗

JPQL

0 ✗

Native Query

0 ✗

Query API

How would you retrieve a single result using JPQL?

22 ✓

0 ✗

`query.findOne();`

2 ✗

`query.getOneResult();`

0 ✗

`query.getSingleResult();` `query.single();`

Which of the following is NOT true about native queries in JPA?

20 ✓



1 ✗

Native queries can return entity objects.

1 ✗

Native queries can be mapped to return custom results.

2 ✗

Native queries take advantage of database-specific features.

When using JPQL, which clause is used to specify the variables that should be returned in the result?

23 ✓



SELECT

1 ✗

RETURN

0 ✗

EXTRACT

0 ✗

TAKE

In a CriteriaQuery, what is used to create the various parts of the SQL query?

23 ✓

0 ✗

EntityManager

1 ✗

QueryBuilder

0 ✗

EntityTransaction

CriteriaBuilder

Quiz leaderboard



Discuss JPA practical task

Next time: Hibernate.

Questions?

